

SOC Analyst Level

Lecture 14: SOC¹ Tools and Daily Operations

How does a SOC Analyst use daily tools and manage their shift?

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| LECTURE OBJECTIVES

- Identify essential daily SOC tools.
- Understand the role of SIEM in daily work.
- Understand the role of EDR in investigations.
- Understand Email Security for Phishing analysis.
- Understand Firewall, Proxy, and DNS Logs.
- Understand the role of Threat Intelligence.
- Understand the Ticketing System.
- Understand Identity and Cloud tools.
- Learn how a SOC analyst starts their day.
- Learn how to review the Queue.
- Learn how to handle the shift.
- Learn how to maintain analysis quality.
- Build a Daily SOC Workflow.
- Execute a full daily operations scenario.

WHY IS THIS LECTURE IMPORTANT?

A SOC Analyst does not work with just one tool. In a real working day, they might use:

SIEM & EDR

Email Security

Firewall & Proxy Logs

DNS Logs

Threat Intelligence

Ticketing System

Identity Tools

Cloud Security Logs

Documentation System

A good analyst doesn't just know the tools; they know WHEN and WHY to use each one.

| COMMON MISTAKES BY BEGINNERS

A beginner might do the following:

- Opens SIEM only and forgets EDR.
- Analyzes Phishing without Email Headers.
- Sees a suspicious IP but skips Threat Intelligence.
- Sees a suspicious Login but ignores MFA or Mailbox Rules.
- Forgets the Ticketing System.
- Does not write Notes.
- Does not know priorities.
- Treats every Alert the same way.

This makes the analysis incomplete.

| THE SOC TOOLCHAIN CONCEPT

SOC Toolchain refers to the suite of tools used by the Security Operations Center.

Every tool provides a piece of the truth:

- SIEM: Provides Events and Alerts.
- EDR: Provides the Process Tree.
- Email Security: Provides email details.
- Firewall: Provides network connections.
- DNS: Provides domain names.
- Identity: Provides logins.
- Ticketing: Provides case management.
- Threat Intel: Provides indicator reputation.

A good investigation connects them all.

| CORE DAILY SOC TOOLS

- SIEM (Centralized Brain)
- EDR/XDR (Endpoint Protection)
- Email Security Gateway
- Firewall Management/Logs
- Proxy/Web Gateway
- DNS Logs Engine
- Threat Intelligence Platforms
- Ticketing/Case Management System
- Identity and Access Management Logs
- Cloud Security Control Logs
- Vulnerability Management Console
- Communication and Documentation Tools

| SIEM OVERVIEW

The **SIEM** is the center for log collection and analysis.

A SOC L1 uses it to:

- Review alerts.
- Search logs.
- Build Timelines.
- Correlate events.
- Write queries.
- Review Incidents.
- Identify Log Sources.
- Extract evidence.

SIEM is usually the starting point, but it is not the endpoint.

| WHAT DOES THE ANALYST DO IN SIEM?

- Read Alert Details.
- Identify Affected User & Host.
- Identify Source IP.
- Set Time Range.
- Search before and after the alert.
- Review Related Events.
- Use Pivots.
- Build Timeline.
- Determine Severity & Confidence.
- Make a Close or Escalate decision.

SIEM USAGE EXAMPLE

Alert: Multiple Failed Logins Followed by Successful Login

Inside the SIEM we search for:

- Failed Logins counts.
- Successful Login matching details.
- Source IP and GeoIP alignment.
- Is the targeted user privileged?
- Is there Event ID 4672 Assigned?
- Is there PowerShell post-login?
- Are there external active connections?
- Did secondary Alerts trigger?

SIEM helps us connect the story.

| THE LIMITS OF SIEM

SIEM alone is not enough.

Some details might not be present, such as:

- The full Process Tree lineage.
- Memory artifacts and process dumps.
- File quarantine details.
- Email attachment sandbox results.
- Full packet analysis capture data.
- Endpoint isolation status control.
- Direct human User confirmation context.

Therefore, we need other specialized tools.

| EDR / XDR DEFINITION

EDR stands for Endpoint Detection and Response. XDR stands for Extended Detection and Response.

EDR helps understand what happened on the host. It maps out:

- Active Processes.
- Parent/Child Process trees.
- Command Line execution arguments.
- Local File Activity logs.
- Local Network Connections.
- Malware Detonation traces.
- User Session Activity on host.
- Device Telemetry Timeline.
- Isolation Control Status.
- Remediation Actions.

| WHEN DO WE USE EDR?

We use EDR when there is:

Malware Alert

Suspicious PowerShell

Office launching script

Unknown process

File execution

C2-like connection

Persistence indicator

Ransomware behavior

Endpoint compromise

EDR is crucial in Endpoint investigations.

| EDR ANALYTICAL CHECKLIST

Inside EDR, we search for:

- What process started the activity?
- What is the Parent Process?
- What is the exact Command Line?
- Was the file executed?
- Who is the logged user?
- What is the folder path?
- Is there a File Hash (SHA256)?
- Are there external outbound connections?
- Are there subsequent child processes?
- Is there a Persistence mechanism?
- Did the same hash appear on other devices?

EDR CONTEXT: OFFICE EXPLOITATION

Case: winword.exe launched powershell.exe -enc

In the EDR we search for:

- The name of the file the user opened.
- Did the file come from an Email location?
- The full encoded Command Line arguments.
- Did PowerShell download an external binary?
- Did it connect to an external malicious address?
- Was a Scheduled Task created?
- Did another process appear after PowerShell?
- Did EDR block or allow the activity?

COMPARISON: SIEM VS. EDR

Tool Architecture	Operational Role
SIEM	Collects and correlates logs from multiple infrastructure sources. Best for macro visibility.
EDR	Provides deep details about the device and process trees. Best for host micro forensics.
SIEM	Usually where the initial Alert triggers.
EDR	Confirms what actually executed on the Endpoint.

A strong investigation uses both together.

| EMAIL SECURITY GATEWAY

Used to inspect corporate email traffic. It helps in:

- Detecting Phishing templates.
- Inspecting malicious links.
- Inspecting attachment binaries.
- Knowing who received the email.
- Knowing who clicked the link.
- Knowing if the email was quarantined.
- Extracting full Email Headers.
- Extracting Sender and Reply-To structures.
- Searching for a Phishing campaign blast radius.

| TRIGGERS FOR EMAIL INVESTIGATION

We use Email Security for:

- Phishing Reports from users.
- Emails with suspicious attachments.
- Suspicious link metrics.
- Credential Phishing.
- Business Email Compromise (BEC).
- Broad email campaigns.
- User clicked malicious link.
- User opened document attachment.
- Email delivered to multiple users.

Any incident starting from an email must pass through Email Security.

| EMAIL ANALYST'S CHECKLIST

When inspecting an email, ask:

- Who is the actual sender?
- What is the Reply-To address?
- Who received the email?
- Did it reach other users?
- Was the link opened?
- Is there an attachment?
- Was the attachment sandboxed?
- Was the link blocked?
- Did SPF/DKIM/DMARC fail?
- Did the user interact with the email?

EMAIL SECURITY EXAMPLE

Subject: Urgent MFA Verification

Sender: security@company-login-verify.example

URL: hxxps://company-login-verify.example/mfa

Inside Email Security, we look for:

- Number of corporate recipients.
- Who clicked the link?
- Is the link blocked now?
- Was the message delivered or quarantined?
- Are there emails with the same Subject?
- Is there an Attachment included?
- Has the sender appeared before?

FIREWALL LOGS ATTRIBUTES

The Firewall logs connections between networks. It gives us:

Source IP	Destination IP	Destination Port
Protocol	Action: Allow/Block	Time
Bytes Sent/Received	Rule Name	Device/Zone

Very useful for C2 tracking, Exfiltration, and Port Scans.

| WHEN TO AUDIT FIREWALL LOGS

We use Firewall Logs for:

- Suspicious external connections.
- C2-like traffic tracking.
- Data Exfiltration validation.
- Port Scan telemetry.
- Connection to a known bad IP.
- Malware attempting to connect.
- Device connecting to a strange country.
- Large data outbound traffic.

The Firewall shows if the connection was Allowed or Blocked.

FIREWALL SCENARIO: BEACONING

Alert: Host 10.0.0.25 connected to 91.200.10.5:443 every 60 seconds.

In the Firewall we search for:

- Is the connection Allowed?
- How many times did it happen?
- When did it start?
- Are there other devices connecting?
- Data packet sizes?
- Is the destination IP or Domain?
- Did it happen after file execution?
- Is the connection ongoing?

| WEB PROXY CONTEXT

Proxy or Web Gateway records web activity. It gives us:

- User Identity.
- Source IP.
- URL & Domain path.
- HTTP Method (GET/POST).
- Web Category.
- Action taken.
- File Download activity.
- User Agent string.
- Bytes count.
- Time metrics.

Useful in Phishing triage, Malware Download, and Data Upload tracking.

| WHEN TO USE PROXY LOGS

We use Proxy Logs when:

- User clicked phishing link.
- File downloaded from web.
- Suspicious domain visited.
- Large file upload.
- Access to cloud storage.
- User browsing suspicious sites.
- Malware download suspected.
- Credential phishing investigation.

Proxy clarifies what the user visited and downloaded.

| DNS LOGS OVERVIEW

DNS Logs record device queries for domain names. They give us:

- Device Asset IP.
- User context (sometimes).
- Queried Domain string.
- Time generated.
- Response IP returned.
- Query Type (A, AAAA, TXT).
- Resolution Result.

DNS is important because much malware starts by querying a Domain before connecting.

| WHEN TO USE DNS LOGS

We use DNS Logs when dealing with:

- Suspicious Domain name.
- Possible C2 beaconing infrastructure.
- Phishing domain validation.
- Malware domain staging.
- DGA (randomized) domains.
- Repeated identical queries.
- Device connecting to external IP after DNS Query.
- Searching for other devices querying the same domain.

DNS helps us understand the attack spread.

| THREAT INTELLIGENCE (TIP)

Threat Intelligence means information about threat indicators, such as:

IP Reputation

Domain Reputation

URL Reputation

File Hash Reputation

Known Malware Family

Known Campaign

Known Phishing Kit

Known C2 Infrastructure

| USING THREAT INTEL EFFECTIVELY

We use Threat Intel when examining:

- External IP reputation.
- Suspicious Domain history.
- Phishing URL strings.
- File Hash signatures.
- Email sender domains.
- Malware detection names.
- C2-like destination parameters.
- IOC seen across multiple devices.

But do not rely on reputation alone. Link it to context and evidence.

TIP CORRELATION EXAMPLE

IP: 45.77.20.10

Threat Intel Result: Malicious (some sources), Associated with phishing (one source), First seen recently.

Analysis Context:

This is supporting evidence, but not enough alone. If it is paired with:

- Encoded PowerShell arguments.
- Repeated dynamic connections.
- Suspicious domain resolutions.

The case becomes much stronger.

| THE TICKETING SYSTEM OVERVIEW

The **Ticketing System** is the case management platform (e.g., ServiceNow, Jira, TheHive, SIEM/SOAR Case Management).

It is used for:

- Opening a Ticket.
- Writing Case Notes.
- Updating Case Status.
- Escalating to T2.
- Assigning the case.
- Tracking SLA countdown.
- Closing the log.
- Saving analysis documentation.

| WHY TICKETING IS IMPORTANT

Ticketing is important because it:

- Prevents lost cases.
- Shows who is working on what.
- Preserves forensic evidence.
- Preserves analyst notes.
- Clarifies the final decision.
- Supports Shift Handover workflows.
- Supports SOC KPI metrics.
- Proves SOC followed correct procedures.

Analysis without a Ticket or Note may be lost.

IDENTITY & IAM TOOLS ATTRIBUTES

Identity Tools manage accounts and login records (Active Directory, Okta, Entra ID).

Examples of what we look for:

Successful Logins

Failed Logins

MFA Events

Password Reset

Account Lockout

New Device

New Country

Session Tokens

Risky Users

Role Changes

Group Membership

| IDENTITY INVESTIGATION SCENARIOS

We use them for:

- Login from New Country.
- Impossible Travel thresholds.
- Failed then Successful Login tracking.
- MFA Fatigue spamming patterns.
- User entered credentials in phishing page.
- Privileged account involvement.
- Suspicious cloud access anomalies.
- New unmanaged device used.
- Mailbox compromise traces.

Identity is the heart of account investigations.

| CLOUD SECURITY TOOLS OVERVIEW

In modern environments, many systems are in the Cloud (AWS, Azure, GCP). Cloud Logs may include:

- Cloud identity sign-ins.
- Admin resource actions.
- Storage bucket access (S3/Blob).
- Virtual Machine activity.
- Role assignments changes.
- Raw API calls.
- Mass file downloads.
- OAuth app grants.
- Mailbox access events.
- Conditional Access outcomes.

| CLOUD-SPECIFIC TRIGGERS

We use Cloud Logs for:

- Compromised cloud account credentials.
- Login from new global location.
- Large download from Cloud Storage layers.
- Unusual Role assignments.
- Suspicious OAuth app grants.
- Malicious Mailbox rules.
- Access to SharePoint/OneDrive.
- Cloud admin configuration changes.

VULNERABILITY MANAGEMENT

These tools display vulnerabilities and unpatched weaknesses. They help in investigations when we need to know:

- Is the system unpatched against the exploit name?
- Does it have a known critical CVE?
- Is the targeted service exposed externally?
- Is the device marked as high-risk?
- Has a vulnerability been logged on it previously?
- Is the server asset critical?

L1 does not turn every Alert into a Vulnerability Assessment. Use them for Context when needed.

| COMMUNICATION CHANNELS

Communication in SOC may occur via: Teams, Slack, Email, Ticket comments, Incident bridge, Phone call (for critical cases).

However, caution is required:

- Do not send sensitive data via unauthorized public channels.
- Do not discuss a sensitive incident in an open public channel.
- Do not share raw IOC links without defanging them.
- Do not bypass official ticket logging.

| DOCUMENTATION & PLAYBOOKS

The SOC reference center. The analyst needs to check:

- Incident Playbooks.
- Escalation Matrix.
- Contact List parameters.
- SLA Definitions.
- Known False Positives lists.
- Approved Scanner IPs.
- IT Change Windows logs.
- Asset Criticality tiering database.

| STARTING THE SOC SHIFT

At the start of the shift, the analyst must execute:

- Read Handover logs.
- Review active open cases.
- Review Critical/High alerts.
- Check expiring SLAs.
- Identify active Incidents.
- Review new unassigned alerts.
- Ensure operational tools are working.
- Know about active changes/maintenance.
- Review Known Issues dashboard.
- Set shift work priorities.

| PRIORITIZED SHIFT START

Mistake: The analyst opens the first random Alert they see and starts working.

Correct approach:

- Review priorities across the board.
- Review Critical tier alerts first.
- Review cases nearing their SLA threshold.
- Review open active escalations.
- Review aggregated Campaigns.
- Review pending information blocks.

Working randomly leads to losing important critical cases.

| THE QUEUE REVIEW TACTIC

The **Queue** is the operational list of active triggers.

When reviewing the Queue:

- Sort by Severity level.
- Sort by oldest arrival time.
- Sort by closest SLA timer.
- Look for matching duplicates.
- Look for campaign traces.
- Look for the same target User or Host.
- Review pending cases status.
- Properly bundle duplicates.
- Escalate high-priority blocks.

| DAILY SOC CHECKLIST

☐ Handover logs reviewed. ☐ Active critical cases validated. ☐ SLA deadline windows verified. ☐ Unassigned alerts triaged. ☐ Escalated cases followed up. ☐ Pending external responses checked. ☐ High-criticality assets monitored. ☐ Threat intel feeds refreshed. ☐ Ticket notes populated. ☐ End-of-shift handover prepared.

| OPERATIONAL DISCIPLINE DURING SHIFT

During the shift:

- Do not leave a case open without logs or updates.
- Do not escalate without factual evidence.
- Do not close a ticket without a verdict reason.
- Do not delay Critical priorities.
- Do not process Low alerts while Highs sit un-triaged.
- Do not copy raw results without context.
- Do not ignore handover logs.
- Do not ignore repeated alerts if entity changes.

| END OF SHIFT OPERATIONS

Before closing out your shift:

- Update all open cases tickets.
- Write clear current status for each Case.
- Define Pending Actions requirements.
- Define the target Owner.
- Define remaining SLA windows.
- Compile a structured Handover log.
- Highlight active Critical or High incidents.
- Highlight any pending external responses.
- Mention any active running Campaigns.

| QUALITY OF DAILY WORK

The operational quality of an L1 analyst is mapped by:

Accuracy of data verification.

Clarity of ticket Notes.

Triage speed on High/Critical.

Zero premature closures.

No evidence-free escalations.

Using the correct tool vector.

Building accurate Timelines.

Updating case states.

Respecting SLA contracts.

Delivering solid Handovers.

| SCENARIO: ONE TOOL IS NOT ENOUGH

SIEM Trigger: Suspicious PowerShell Execution

If you use the SIEM dashboard alone, you only view: Time, Account, Machine name, raw string.

To resolve the case, you need more:

- EDR: To map out the complete Parent Process tree.
- Email Security: To see if it was delivered via an active attachment file.
- DNS Logs: To verify if a domain resolution accompanied the call.
- Firewall Logs: To check if a network connection succeeded.
- Threat Intel: To review target IP destination reputation.
- Ticketing: To preserve the investigative notes.

WORKFLOW: SUSPICIOUS POWERSHELL

Operational Triage Steps:

- 1 SIEM: Read the incoming Alert. Extract User, Host asset, and Time.
- 2 EDR: Pivot to endpoint; review full Process Tree context.
- 3 EDR: Analyze and decode the Command Line string arguments.
- 4 Email Security: Cross-verify if a file write occurred from a mail attachment.
- 5 DNS Logs: Check if a domain lookup occurred right around the timestamp.
- 6 Firewall Logs: Verify if an outbound session was Allowed or Blocked.
- 7 Threat Intel: Check target external destinations reputation.
- 8 Ticketing: Document all findings into a cohesive timeline note, then close or escalate.

WORKFLOW: PHISHING TRIAGE

Operational Triage Steps:

- 1 **Email Security:** Inspect Sender envelope, Reply-To header, embedded URLs, and Hashes.
- 2 **Email Security:** Identify all delivery recipients within the enterprise.
- 3 **Proxy:** Verify if any host proxy session logs confirm a click/connection to the URL.
- 4 **Identity:** Check for anomalous successful logins following the interaction timestamp.
- 5 **Mailbox Logs:** Search for newly established Mailbox Forwarding Rules.
- 6 **EDR:** Audit host logs to see if a downloaded document ran a child process.
- 7 **SIEM:** Synthesize all logs into a master chronological incident timeline.
- 8 **Ticketing:** Document the case findings and escalate if credential theft or execution is confirmed.

WORKFLOW: ACCOUNT COMPROMISE

Operational Triage Steps:

- 1 Identity Logs: Audit Successful vs. Failed Logins across the timeframe.
- 2 SIEM: Execute a broad lookup for the external Source IP across all company identities.
- 3 MFA Logs: Check challenge-response patterns, looking for push exhaustion.
- 4 User Contact: Validate the activity directly with the employee per company protocol.
- 5 Cloud Logs: Audit cloud storage downloads, application grants, or permission adjustments.
- 6 Mailbox Logs: Verify inbox rule configurations.
- 7 EDR: Inspect the host device parameters used during authentication.
- 8 Threat Intel: Verify Source IP malicious history or Tor/VPN proxy mapping, then document and remediate.

WORKFLOW: MALWARE EXECUTION

Operational Triage Steps:

- 1 EDR: Analyze the specific Detection signature and rule matching details.
- 2 EDR: Verify the enforcement outcome: Was the file Blocked or successfully Executed?
- 3 EDR: Extract the file Name, absolute Path, and cryptographic Hash signature.
- 4 EDR: Map out the full Process Tree lineage leading to the trigger.
- 5 Firewall/DNS: Look for outbound connections or domains resolved by the process.
6. Email/Proxy: Track the initial download source vector (how did it arrive?).
7. SIEM: Run an enterprise-wide search for the same file hash across all assets.
8. Threat Intel: Verify the hash reputation, log findings, and isolate host if execution is true.

WORKFLOW: DATA EXFILTRATION

Operational Triage Steps:

- 1 SIEM: Read the anomalous volume transfer alert payload data.
- 2 Firewall/Proxy: Extract the total data size transferred and target destination IP/URL.
- 3 Identity: Pinpoint the exact User Account running the outbound session.
- 4 Endpoint: Map the physical host machine ID driving the workflow.
- 5 File Logs: Audit file modification and archiving logs right before the transfer window.
6. Cloud Logs: Check if data was aggregated from corporate cloud storage levels.
7. Threat Intel: Analyze destination IP reputation and escalate if size is high or content is critical.

STRATEGIC TOOL SELECTION MATRIX

· Manage case documentation, notes history, and SLA countdowns

Investigative Target	Primary Tool Solution Feed
Determine authentication origin and success status	SIEM / Identity Provider Logs
Analyze local host process creation trees and line arguments	EDR Console
Track message delivery paths, recipients, and attachments sandboxing	Email Security Gateway
Confirm web page arrival following a link click event	Secure Web Proxy / Web Gateway
Verify if a network session was Allowed or Blocked at network edge	Firewall Session Logs
Identify host-level domain queries and name resolutions	DNS Logs engine
Audit external reputation records for an unknown indicator	Threat Intelligence Platforms
Ticketing System	

| COMMON TOOL SELECTION MISTAKES

Avoid these investigative blind spots:

- Attempting to triage malware using only high-level SIEM logs without looking at EDR trees.
- Investigating phishing reports without checking email gateway metrics or message headers.
- Triaging a compromised account alert without looking at IAM or Identity Provider logs.
- Analyzing command-and-control behavior without pulling network firewall or DNS query logs.
- Assessing potential data exfiltration threats without logging the exact byte size transferred.
- Escalating an incident to higher tiers without generating an official work ticket.
- Closing alerts without checking corporate white-lists or approved maintenance windows.

| HANDLING KNOWN FALSE POSITIVES

A **Known False Positive** is a baseline, legitimate corporate action that matches an alert logic but poses zero risk.

Examples of common benign noise:

- Authorized vulnerability scanners running automated network sweeps.
- Corporate IT deployment software pushing system changes via PowerShell.
- Enterprise backup agents performing scheduled mass data transfers.
- Network monitoring tools running persistent service checks.
- System admins executing local server automation scripts.
- Authorized penetration testing windows conducted by security teams.

Never close an alert instantly. Verify that the time, account, host, and activity strictly align with the approved record.

| APPROVED LISTS HYGIENE

The SOC relies on continuously updated databases to accurately filter out standard operational noise:

Scanner Profiles

Approved Scanner IP targets.

Admin Tooling

Authorized Admin tools/scripts.

Web Perimeters

White-listed domains/CDN paths.

Cloud Routing

Approved corporate Cloud ranges.

System Access

Authorized Service Accounts.

Change Management

Active maintenance Windows.

NAVIGATING CHANGE WINDOWS

A **Change Window** is a pre-approved slot for IT infrastructure maintenance or updates.

Example: Systems engineering pushes software patches every Friday between 10:00 PM and 01:00 AM.

If alerts like mass reboots, process spawns, or internal scans trigger during this specific window, they may be legitimate.

However, an L1 analyst must explicitly verify:

- Is the observed activity part of the logged change request?
- Are the affected hosts inside the approved maintenance scope?
- Does the calling process map to a valid service account?
- Is official approval documented in the change ticket?
- Are there secondary, anomalous indicators that mismatch the change profile?

ASSET CRITICALITY TIERING

Not all devices on the network hold equal risk or business value. Alerts must be prioritized based on system criticality:

Criticality Tier	Target Core Asset Profiles	SOC Action Strategy
Tier 0 (Critical)	Domain Controllers, Identity Identity Providers, PKI Servers.	Immediate critical priority triage response regardless of alert severity.
Tier 1 (High)	Core Database Servers, Central File Share Servers, Executive (VIP) Endpoints.	High priority response; escalate severity tiering instantly.
Tier 2 (Medium)	Standard Production Workstations, Internal Application Servers.	Standard SLA timeline triage path execution.
Tier 3 (Low)	Isolated Test Lab Machines, Guest Network VLAN Devices.	Lower operational response path priority assignment.

| USER CRITICALITY MAPPING

Similarly, the privilege level of the targeted corporate identity directly multi-folds the potential risk of an alert:

Domain Admins

Full Active Directory control potential.

Global Cloud Admins

Full enterprise cloud tenant footprint compromise.

C-Level / VIPs

Target for high-value espionage and spearphishing.

Finance Controllers

Direct exposure to wire transfers and monetary exfiltration.

Core Service Accounts

High network privilege levels; often bypass standard MFA blocks.

| DAILY QUALITY CHECKLIST PER ALERT

Before closing out any ticket or passing it up to Tier 2, ensure you answer every point on this quality checklist:

- ☒ Do I fundamentally understand what triggered this alert logic?
- ☒ Have I explicitly identified the affected User identity?
- ☒ Have I explicitly isolated the precise Host machine asset?
- ☒ Have I defined the exact timeframe of the anomaly?
- ☒ Have I mapped all core network points (Source and Dest)?
- ☒ Have I checked for secondary, correlated logs in the SIEM?
- ☒ Did I pivot to the most appropriate specialized tool?
- ☒ Are my Case Notes fully structured, clean, and reproducible?
- ☒ Is my assessment of Severity and Confidence mappers mounded in logs?
- ☒ Is the final case disposition fully justified with zero guesswork?

PRACTICAL EXERCISE 1: TOOL SELECTION - PHISHING

Scenario: A user clicks an embedded link inside a suspicious email message framework.

Map the optimal investigative tool path to resolve this case:

- **Email Security Gateway:** Extract the URL string, parse headers, and map total corporate recipients.
- **Secure Web Proxy Logs:** Confirm whether a network session actually reached the destination landing page.
- **Identity Provider Logs:** Check for sudden successful cloud authentications right after the click.
- **Mailbox Audit Logs:** Audit account settings to ensure no hidden mailbox forward rules were added.
- **SIEM Engine:** Link all these distinct logs into a comprehensive chronological timeline.
- **Ticketing Tool:** Formally preserve all case findings, IOCs, and decisions.

PRACTICAL EXERCISE 2: TOOL SELECTION - MALWARE

Scenario: The SIEM registers an EDR alert flagging a malicious file detection on host HR-LAPTOP-07.

Map the optimal investigative tool path to resolve this case:

- **EDR Console:** Pinpoint the file Name, absolute Path, cryptographic Hash, tree lineage, and block status.
- **Threat Intelligence Platforms:** Upload the file hash to identify its malware family and known bad network targets.
- **DNS Logs:** Search network-wide to see if any other local asset resolved the malicious C2 domain names.
- **Firewall Session Logs:** Confirm whether any outbound C2 connection was Allowed or successfully Blocked.
- **Email Gateway / Proxy:** Reconstruct how the malicious binary originally landed on the endpoint.
- **Ticketing Tool:** Document the full case story, append hashes, and log the escalation handover.

PRACTICAL EXERCISE 3: SHIFT START PRIORITY

You log into your SOC shift at 08:00 AM and find the following items waiting in your workspace queue:

- 10 isolated Low-severity alerts. | An active Phishing Campaign case container.
- A High-severity escalation case pending Tier 2 forensic review.
- A High-severity incident whose SLA response timer breaches in exactly 20 minutes.
- A brand new, unassigned Critical-severity alert trace.

Organize your task execution order based on risk and SLA contracts:

1. Triage the brand new Critical-severity alert immediately (highest threat signature).
2. Process the High-severity incident nearing its SLA breach window (must resolve within 20 mins).
3. Follow up with Tier 2 regarding the pending forensic escalation case.
4. Review and update the active multi-user Phishing Campaign.
5. Process the 10 Low-severity background alerts last.

PRACTICAL EXERCISE 4: END OF SHIFT HANDOVER

Open Active Case: Potential Account Compromise

Current State: Escalated to Tier 2 / Identity Management

Pending Items: Employee password verification and Tier 2 deep endpoint forensic audit.

Collected Evidence: Successful web login from anomalous country + malicious forward rule compiled.

Task: Draft a concise, high-quality Handover entry to transfer this case to the incoming shift team.

HANDOVER ANSWER KEY

The formal handover entry for your case tracking log should be structured as follows:

```
Handover Log Entry - Case #SOC-2026-8941 - Title: Potential Account Compromise - Identity Triage - Status: Escalated (Pending External Verification) - Current State: Confirmed a successful authentication from an anomalous foreign country, followed immediately by the creation of an unauthorized mailbox forwarding rule. - Pending: Awaiting direct employee verification via phone and Tier 2 endpoint process tree analysis. - Target Owner: Incoming Shift Analyst (Tier 1 Queue) - SLA Context: High priority tier; next operational update or follow-up required within 30 minutes. - Next Steps: 1. Follow up with user to confirm location. 2. Verify with IAM if password reset and active session revocation were successfully completed. 3. Monitor SIEM for secondary login attempts from the anomalous IP.
```


MASTER TRAINING SCENARIO

Analyze and process this complex, multi-tool security incident trace:

- SIEM Feed: Alerts trigger flagging "Suspicious PowerShell Execution on host FIN-LAPTOP-03". - EDR Forensics: Process logs confirm excel.exe launched powershell.exe passing an encoded Base64 argument. - Email Gateway: Audit logs verify the user finance.manager received an inbound message containing "salary_update.xlsx" from an external public Gmail address right before the trigger. - DNS Ingestion: Telemetry records show the host ran a DNS resolution call targeting domain "salary-check-update.example". - Firewall Sessions: Network logs confirm sustained outbound sessions connecting to external IP 91.200.10.5 on port 443 occurring consistently every 60 seconds. - Threat Intelligence: Reputational databases flag the target IP and domain as highly suspicious and linked to malware infrastructure. - Case Queue: The ticket is currently unassigned (New) with no analyst notes logged. The High-severity SLA countdown breaches in exactly 25 minutes.

| SCENARIO ANSWER: TOOL ORCHESTRATION

To investigate this case fully, map out how your toolchain integrates to track the threat:

- **SIEM Engine:** Used to ingest the initial trigger alert and pull the contextual entity identifiers (User, Host, Timestamps).
- **EDR Console:** Used to pull the process tree, pinpoint Excel as the parent, and extract the raw encoded PowerShell string.
- **Email Gateway:** Used to trace the inbound phishing delivery path, extract the sender address, and map other potential internal targets.
- **DNS Logs:** Used to verify the domain name lookup and search network-wide for secondary infections.
- **Firewall Logs:** Used to validate that the outbound session was Allowed and track byte transfer volume.
- **Threat Intel Feeds:** Used to verify indicator reputation to support your escalation justification.
- **Ticketing Platform:** Used to log notes and pass the case up before the SLA expires.

| SCENARIO ANSWER: RECONSTRUCTED TIMELINE

Your report must map out the exact progression of the attack chronologically:

```
Chronological Incident Timeline: 1. [Timestamp T1] - Inbound phishing email containing "salary_update.xlsx" delivered to finance.manager (Email Gateway). 2. [Timestamp T2] - User opens the document attachment and enables malicious macro macros (EDR Telemetry). 3. [Timestamp T3] - The parent excel.exe process executes an encoded child powershell.exe script call (Host Process Audit). 4. [Timestamp T4] - Host requests network name resolution for domain "salary-check-update.example" (DNS Server Logs). 5. [Timestamp T5] - Host establishes a persistent outbound connection channel to 91.200.10.5 on port 443 cycling every 60 seconds (Firewall Logs). 6. [Timestamp T6] - Threat Intelligence databases flag the external domain and destination IP as malicious infrastructure (TIP Feeds). 7. [Timestamp T7] - SIEM correlates logs and triggers the primary "Suspicious PowerShell" triage alert for the SOC queue.
```

| COMMON MISTAKES IN DAILY OPS

Avoid these critical operational errors during your shift:

- Relying exclusively on a single tool or dashboard window without pivoting to confirm activity.
- Failing to log actions and evidence inside the formal Ticketing System.
- Ignoring SLA timers and allowing high-priority tickets to breach their response windows.
- Skipping the Shift Handover logs at the start of your shift, leading to duplicated investigations.
- Leaving your active cases completely silent without clear, periodic log updates.
- Mass-closing active alerts without triage simply because the queue volume is high.
- Escalating unverified alerts to Tier 2 without providing clear technical logs or summaries.
- Failing to cross-check alerts against Known False Positive databases or corporate white-lists.
- Skipping Threat Intelligence reputation checks for external IPs or domains.
- Clocking out at the end of a shift without passing over a structured, clear handover record.

THE GOLDEN RULE OF TOOLCHAIN CORRELATION



"A single tool is never enough to resolve an alert. High-fidelity threat analysis requires orchestrating your entire toolchain to connect separate data points into a single truth."

The SIEM flags the event → EDR forensic trees analyze execution → Email gateways uncover delivery origins → Firewall/DNS logs map perimeter sessions → Threat Intel provides context → Ticketing tracks the work → Handovers preserve operational continuity.

| LECTURE SUMMARY

- The Daily SOC Toolchain: Mechanics of SIEM, EDR, Email Gateways, Firewalls, Proxies, and DNS.
- Orchestrating tools to ask specific investigative questions without data siloing.
- Shift Start Routines: Handover ingestion, queue hygiene, and risk prioritization.
- Operational Discipline: Managing SLA countdowns and updating active case notes.
- End of Shift Handover: Formulating clear technical summaries for incoming teams.
- Filtering Noise: Handling Known False Positives lists and navigating Change Windows.
- Contextual Triaging: Aligning alert response paths with Asset Criticality and User Privileges.
- Reconstructing multi-vector attack scenarios from raw data points into structured reports.

REVIEW QUESTIONS

- 1 What is the fundamental analytical difference between a SIEM log and EDR process telemetry?
 - 2 Under what specific threat conditions must an analyst pivot into the Email Security Gateway console?
 - 3 Why do perimeter network firewall logs provide definitive proof compared to local application alerts?
 - 4 Explain why DNS query logs are a premier tool for discovering the network blast radius of malware.
 - 5 What is the role of a Threat Intelligence platform during an active IP investigation?
 - 6 Why is a ticketing system considered an essential security preservation system, not just an admin utility?
- 1 How do IAM (Identity logs) assist in triaging an Impossible Travel alert?
 - 2 Detail the exact step-by-step routine a professional analyst performs during the first 30 minutes of a shift.
 - 3 What specific data criteria must be included in a professional Shift Handover log?
 - 4 Why can't an analyst automatically close an alert simply because it matches a "Known False Positive" profile?
 - 5 How does an active Change Window impact your assessment of an administrative PowerShell execution alert?
 - 6 Explain how Asset Criticality multi-folds your triage priority for an identical privilege assignment alert.

| KNOWLEDGE ASSESSMENT QUIZ

1. An EDR console is primarily leveraged by an L1 analyst to investigate:

A) Perimeter traffic routing | B) **Host-level process trees, parent/child relationships, and command-line execution lines** | C) Network firewall rule sets | D) Inbound email metadata.

2. Which specialized tool must be consulted to map the total enterprise recipients of a malicious attachment?

A) Web Proxy | B) Active Directory logs | C) **Email Security Gateway** | D) Vulnerability scanner records.

3. Network firewall logs are uniquely positioned to prove whether a connection was:

A) Encrypted using standard SSL | B) **Explicitly Allowed or Blocked by security policy at the perimeter edge** | C) Initiated by a Word macro | D) Associated with a specific file hash signature.

4. Threat Intelligence platforms provide analysts with:

A) Raw event log storage | B) **Reputational context, malware classification, and infrastructure history for an external indicator** | C) Automated host isolation actions | D) Direct user account password management controls.

5. The very first operational step an analyst executes upon starting a shift is:

A) Mass-closing the lowest-severity queue items | B) **Reviewing the outgoing shift Handover log and auditing critical pending cases** | C) Writing custom KQL queries | D) Isolating a production server asset.

| HOMEWORK: THE SHIFT CHALLENGE

Triage and build a formal case response report based on this raw daily operations data block:

- Target Profile: User account "finance.manager" | Machine asset: "FIN-LAPTOP-03" - SIEM Feed: Triggers a High-severity "Suspicious PowerShell Execution" alert. - Endpoint EDR: Telemetry confirms excel.exe spawned powershell.exe passing an encoded argument string. - Email Gateway: Audit logs record an inbound message delivery containing file "salary_update.xlsm" right before the spawn. - Enterprise DNS: Host logs catch a name resolution lookup targeting domain "salary-check-update.example". - Edge Firewall: Session data logs confirm persistent connections routing traffic to external IP 91.200.10.5:443 every 60 seconds. - Threat Intelligence: Reputation databases flag the external IP and domain as malicious malware staging infrastructure. - Case Queue: The ticket is currently unassigned (New) with zero notes. The High-priority SLA countdown expires in exactly 30 minutes.

Deliverables: List required tools, construct the chronological timeline, state your calculated Severity/Confidence, write a professional Case Note, write an Escalation Message, draft the Shift Handover log, and state your final decision.

| HOMEWORK ANSWER TEMPLATE

- SOC Operations Case Template:
- Primary Incident Alert Name:
- Affected Target Entities (User/Host):
- List of Utilized Toolchain Ecosystem Apps:
- Chronological Incident Timeline Table:
- Extracted Technical Key Evidence:
- Calculated Case Severity Tier & Justification:
- Analysis Confidence Level & Reasoning:
- Structured Professional Case Note Document:
- Formal Technical Escalation Message:
- Appended Shift Handover Summary Entry:
- Final Analyst Disposition Decision:
- List of Recommended Containment Actions:

| HOMEWORK BRIEF ANSWER KEY

Orchestrated Toolchain Suite: SIEM (Alert triage), EDR (Process trees), Email Gateway (Delivery source), DNS Logs (Domain scoping), Firewall (Connection verification), Threat Intel (Reputation history), Ticketing System (Case preservation).

Severity Mapping: High | **Confidence Level:** High | **Final Decision:** Escalate Immediately

Justification Statement: The logs demonstrate a verified multi-vector attack chain on a highly privileged account. Excel disarmed macro security to spawn an encoded shell, which resolved an intelligence-flagged malicious domain and established a persistent command-and-control connection pattern at the network edge. This requires immediate escalation and host isolation.

| HOMEWORK KEY: CASE NOTE DOCUMENT

The structured, professional case note appended to the ticket log must read as follows:

```
Case Investigation Record - Ticket #INC-2026-9042 - Analyst: [Your Name] | Time of Review: UTC Timestamp -  
Scope: Audited multi-source logs across SIEM, EDR, Email Sec, DNS, and Firewall platforms for host FIN-LAPTOP-03. -  
Findings: Verified inbound delivery of phishing file "salary_update.xlsx" to user finance.manager. Telemetry confirms  
excel.exe spawned powershell.exe executing an encoded string payload. Host subsequently executed a DNS lookup for  
domain "salary-check-update.example" and opened persistent, cycling outbound connections to external IP 91.200.10.5  
on port 443 every 60 seconds. Threat Intelligence databases confirm the IP and domain are flagged malicious C2  
infrastructure. - Verdict: Confirmed phishing exploit leading to malicious process execution and active command-and-  
control beaconing. - Next Steps: Escalating case immediately to Tier 2 / Incident Response. Recommended containment:  
Enforce EDR host isolation protocols, block the malicious IP/Domain at the firewalls, and execute a cloud session  
revocation with forced password reset for the compromised identity.
```

What's Next?

Lecture 15: Capstone Practical Case

The Final Comprehensive Triage

Simulation

You will operate a simulated live SOC shift environment. You must independently triage multiple overlapping alerts, write complex queries, orchestrate your toolchain, construct timelines, map behaviors to MITRE ATT&CK, assign accurate severity metrics, and compose a formal incident report from scratch.



INSTRUCTOR GUIDE: SUGGESTED TIME ALLOCATION

Suggested Time Allocation Plan (3 Hour Total Session):

10 Minutes

Time Block	Target Instructional Activity Focus
15 Minutes	Review case management workflows, SLAs, and handovers from Lecture 13.
25 Minutes	Introduce the SOC Toolchain concept and the "Layers of Truth" methodology.
25 Minutes	Deep dive into the operational interface differences between SIEM and EDR.
25 Minutes	Map Email Security Gateway triage steps and header extraction paths.
25 Minutes	Analyze network log sources: Perimeter firewalls vs. Proxy web gateways vs. DNS lookup engines.
20 Minutes	Calibrate how to leverage Threat Intelligence reputation feeds objectively without guesswork.
20 Minutes	Deconstruct Identity logs (IAM), Cloud posture feeds, and Ticketing systems tracking.
25 Minutes	Break down Daily SOC Workflows: Shift Start checklists, alert queue flows, and incident response.

| INSTRUCTOR GUIDE: FOCUS POINTS & KEY CONCEPTS

Core Pedagogical Directives for the Instructor:

- Break the student habit of "SIEM-only tunnel vision." Stress heavily that a SIEM generates the alert flag, but specialized platforms (EDR, Gateways) provide the micro forensic truth.
- Train students to select tools based on an explicit, structured question (e.g., "I need to check process lineages, therefore I open the EDR console").
- Enforce absolute accuracy when documenting Case Notes. Do not accept loose descriptions; make students use the "Scope, Evidence, Verdict, Next Step" grid format.
- Emphasize that an indicator's clean reputation in Threat Intel does not mean it is safe. Zero-days or newly registered attack domains will show zero hits initially. Context is king.
- Make the Shift Handover exercise highly interactive. Handovers are a top point of operational failure in real-world SOC environments.
- Clearly demonstrate how Asset Criticality and User Privilege metrics can multiply the priority level of an otherwise standard event.
- Ensure students practice defanging URLs in their ticket documentation to avoid accidental clicks within the platform architecture.

INSTRUCTOR GUIDE: EXPECTED STUDENT MISTAKES

Mistake 1: Analyzing Malware via SIEM Logs Only

The student attempts to fully triage a malware alert strictly using parsed SIEM text lines, completely skipping process lineage audits.

Correction: Force them to pivot to the EDR view. Show them that only an EDR charts the parent/child process tree mapping required to identify the root file component.

Mistake 2: Blind Tool Selection

The student tries to find email gateway delivery lists using web proxy logs, or checks network file sizes using DNS server resolution logs.

Correction: Make them define their goal before opening a tool. "If you need file download origins, you open the Proxy or Mail logs. If you need name resolutions, you open DNS logs."

Mistake 3: Zero Documentation In Ticketing

The student solves the technical puzzle across the tools but leaves the ticket entirely blank or provides a 1-word note.

Correction: Remind them that in an enterprise SOC, any analysis that isn't formally documented in the ticket does

Mistake 4: Automated Whitelist Closure

The student sees an administrative script running during a Change Window and instantly closes the ticket as benign without tracking the calling account.

Correction: Attackers intentionally use Change Windows and masquerade as admin scripts (Living off the Land).

CLOSING MESSAGE TO STUDENTS

"A professional analyst is defined by their ability to orchestrate the entire toolchain. Do not look at tools as separate tabs; look at them as a unified ecosystem designed to unmask the threat."

Master your pipeline, document your steps, protect your timestamps, and your value to the security operations team will be absolute.